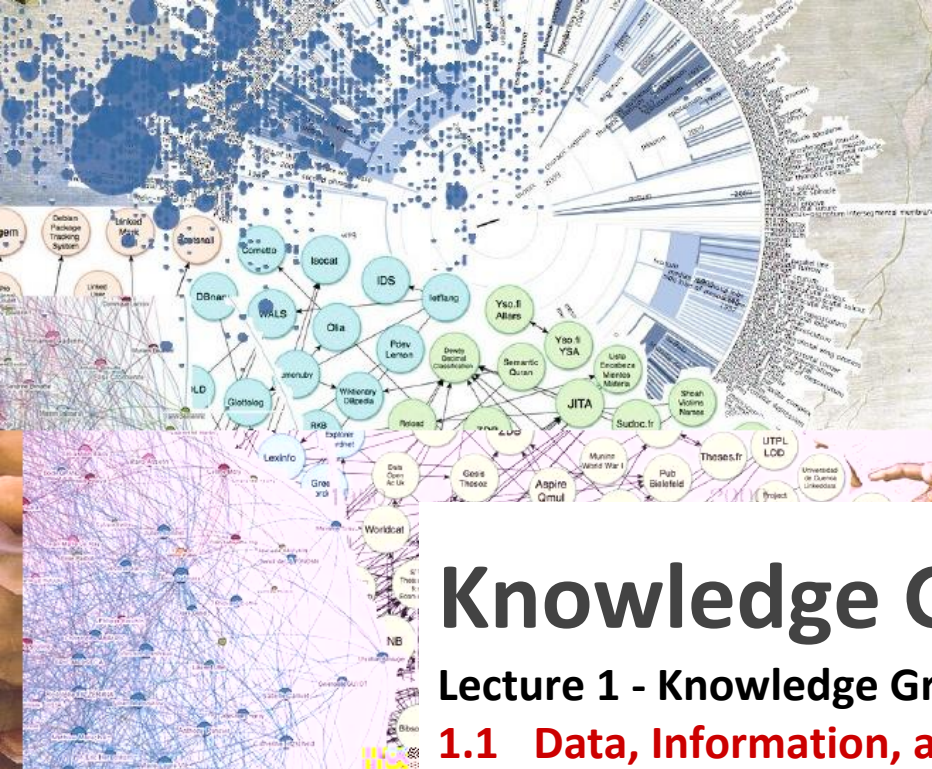




Knowledge Graphs

Lecture 1 - Knowledge Graphs in the Web of Data

1.1 Data, Information, and Knowledge

Prof. Dr. Harald Sack & Dr. Mehwish Alam

FIZ Karlsruhe - Leibniz Institute for Information Infrastructure

AIFB - Karlsruhe Institute of Technology

Autumn 2020



FIZ Karlsruhe

Leibniz-Institut für Informationsinfrastruktur

Knowledge Graphs

Lecture 1: Knowledge Graphs in the Web of Data

1.1 Data, Information, and Knowledge

1.2 How to Represent Knowledge?

1.3 The Art of Understanding

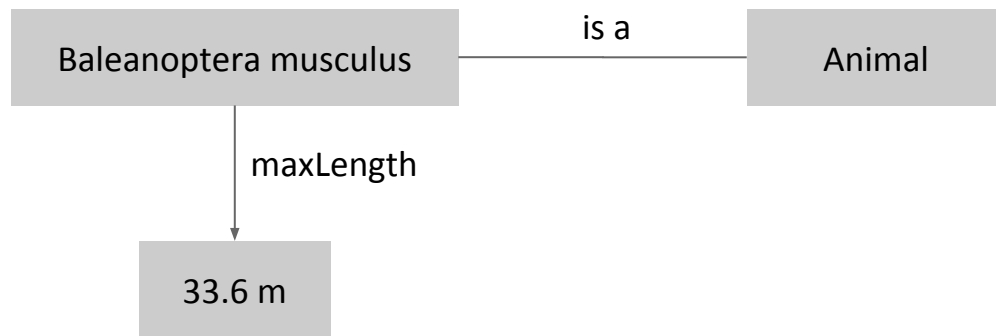
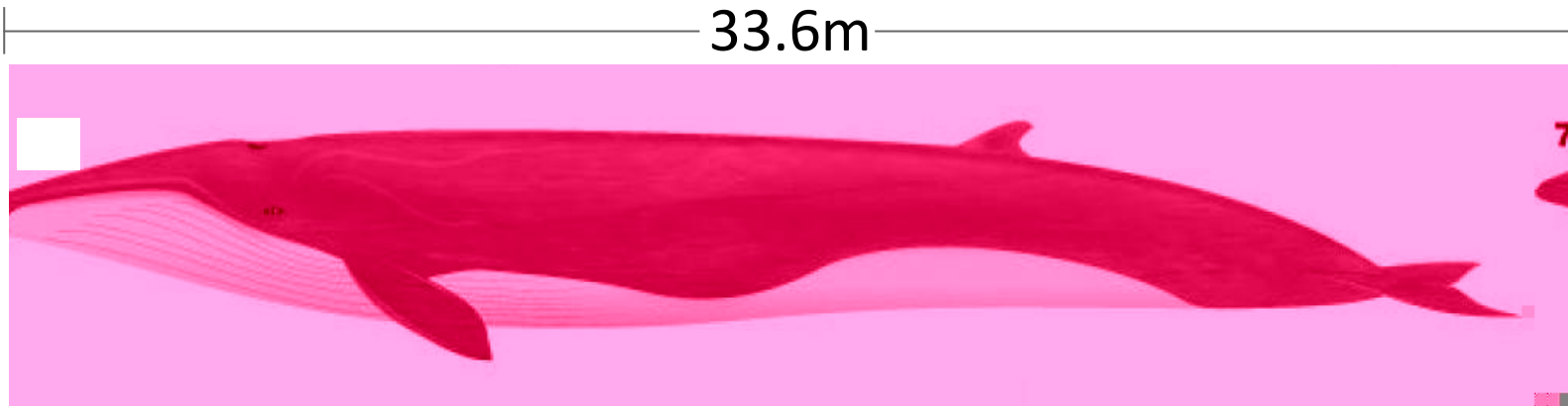
1.4 Towards a Universal Knowledge Representation

1.5 The Semantic Web

1.6 Linked Data and the Web of Data

33.6

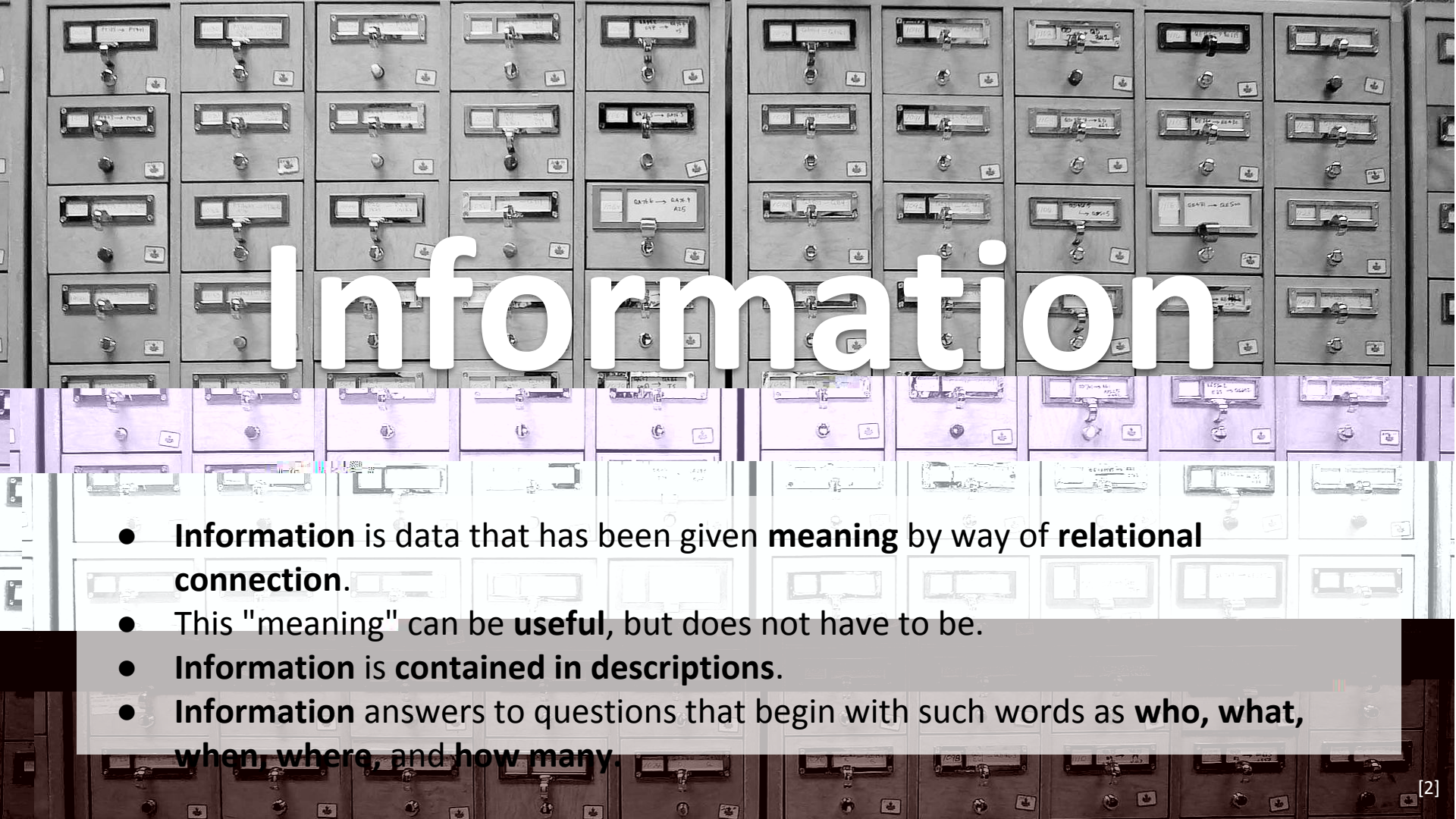
33.6 m



`BaleanopteraMusculus` $\sqsubseteq$ `Animal` $\sqcap \forall \text{maxLength}.\leq 33.6$

Data

- Data is raw.
- It simply exists and has no significance beyond its existence (in and of itself).
- It can exist in any form, usable or not.



Information

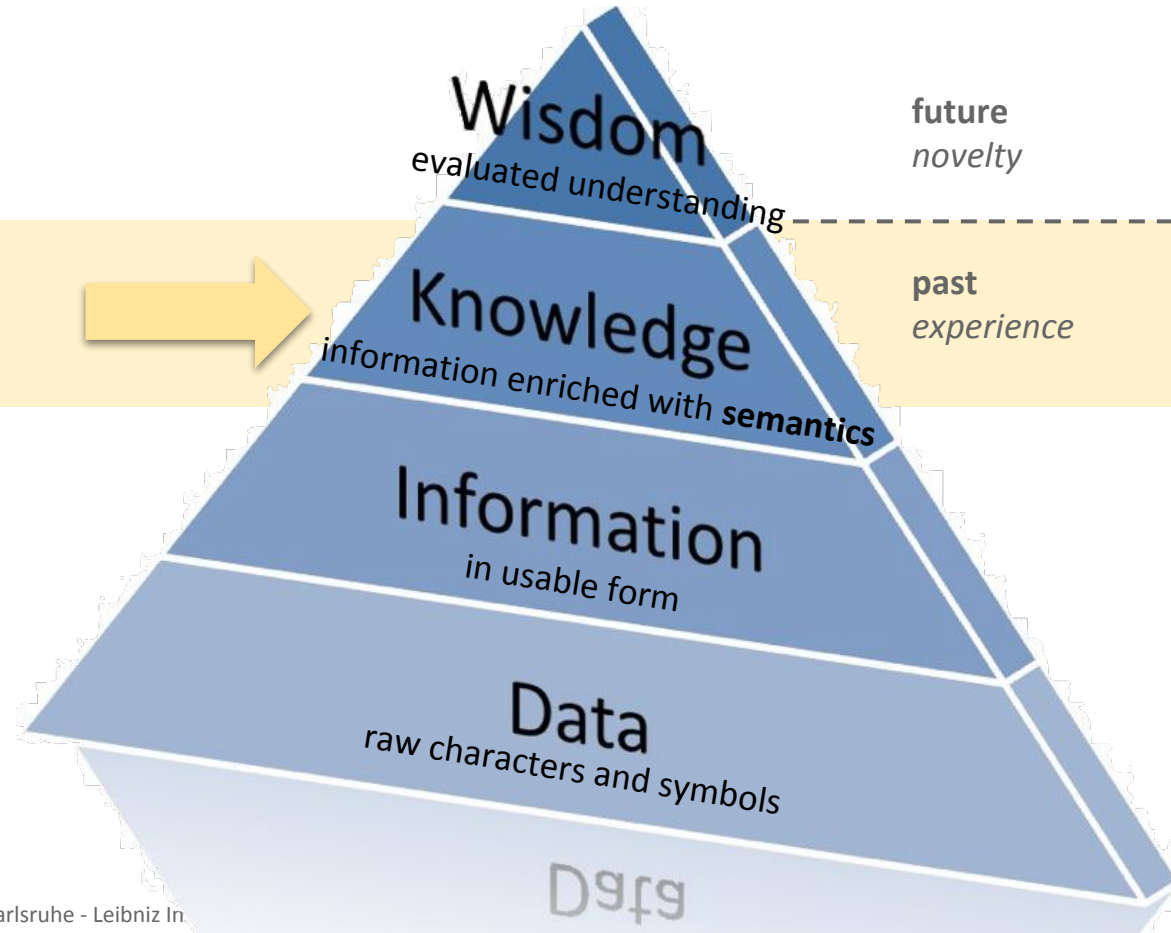
- **Information** is data that has been given **meaning** by way of **relational connection**.
- This "meaning" can be **useful**, but does not have to be.
- **Information** is **contained in descriptions**.
- **Information** answers to questions that begin with such words as **who, what, when, where, and how many**.

Knowledge

- **Knowledge** is the appropriate collection of information, such that it's intent is to be **useful**.
- **Wisdom** is the ability to make **sound judgments** and **decisions**.
- **Understanding** is a continuum that leads from **data**, through **information** and **knowledge**, and ultimately to **wisdom**.

Data transforms to **information** by *convention*, **information** to **knowledge** by *cognition*, and **knowledge** to **wisdom** by *contemplation*.

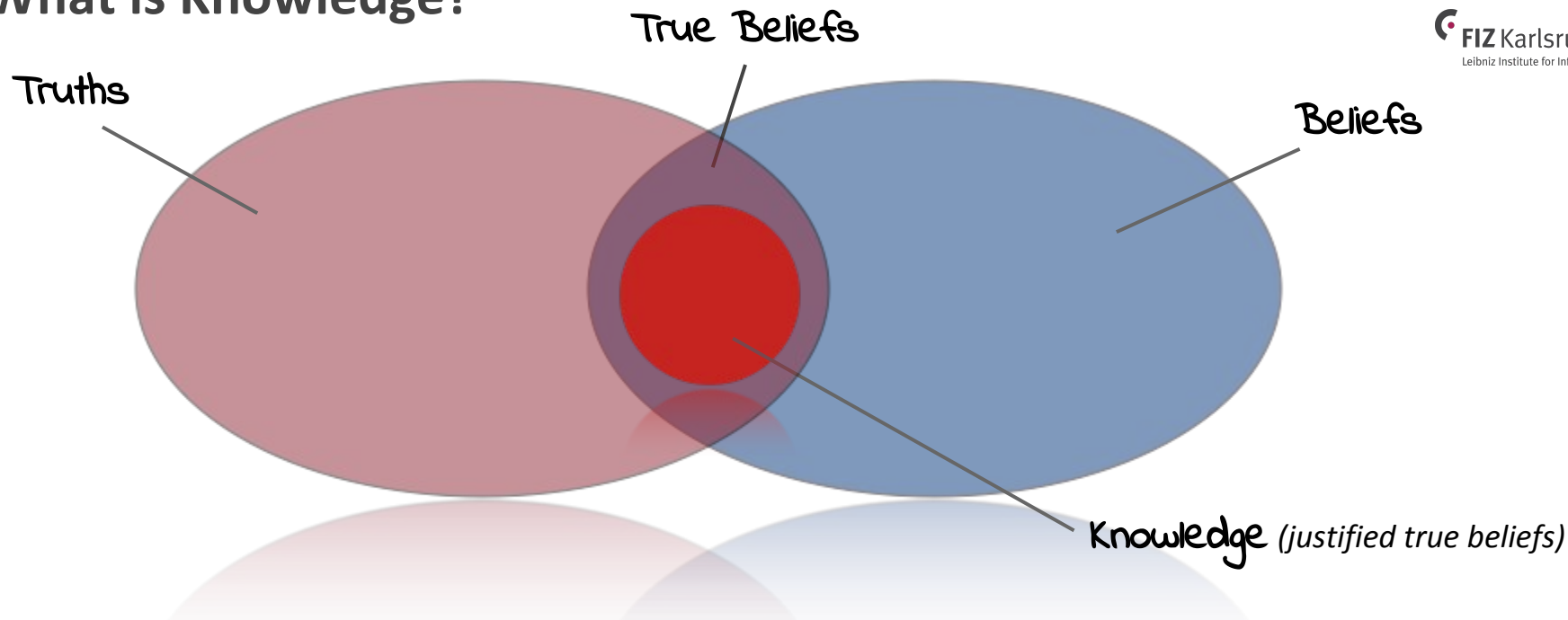
From Data to Knowledge



DIKW Pyramid, Ackoff 1989

What is Knowledge?

What is Knowledge?

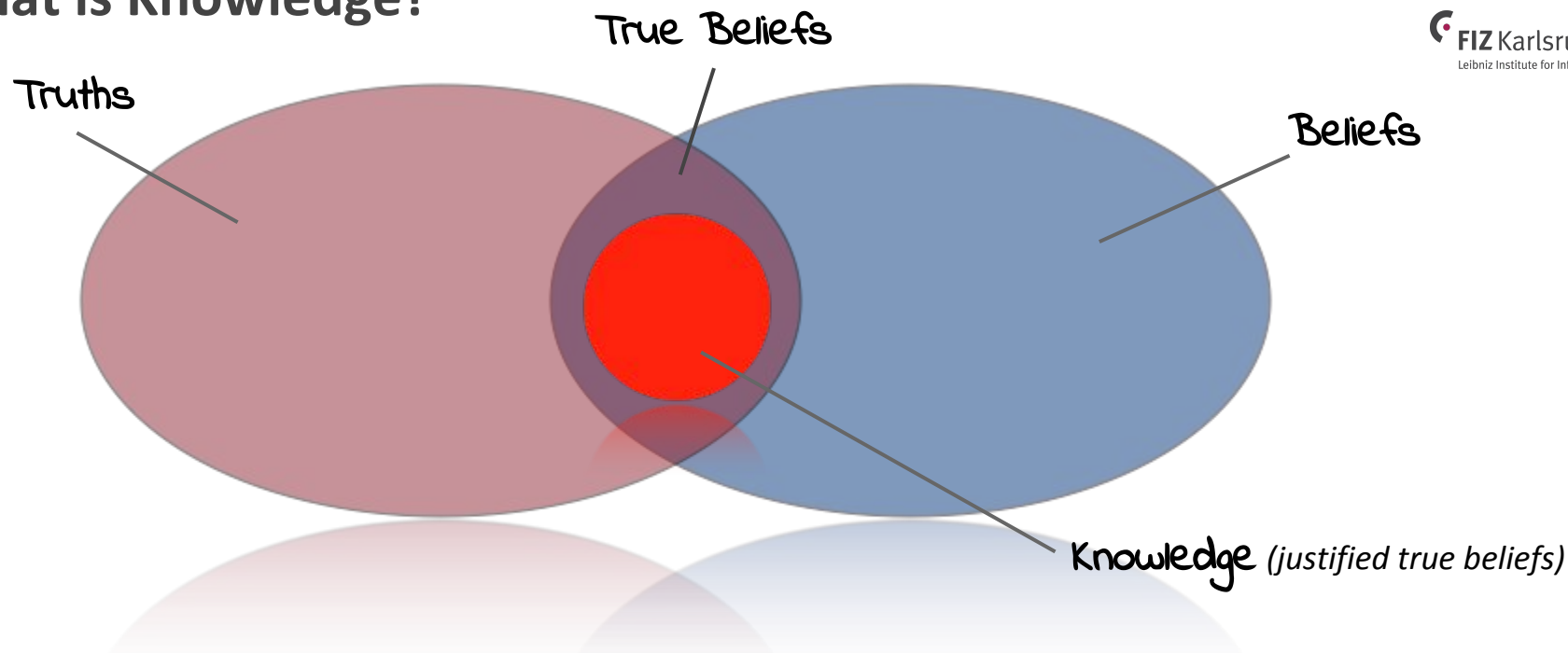


The Tripartite **Analysis of Knowledge**: S knows that p iff

- p is true;
- S believes that p ;
- S is justified in believing that p .

[The Analysis of Knowledge](#), Stanford Encyclopedia of Philosophy, 2001.

What is Knowledge?



Traditional Definition: „*Knowledge is a justified subset of all true beliefs*“

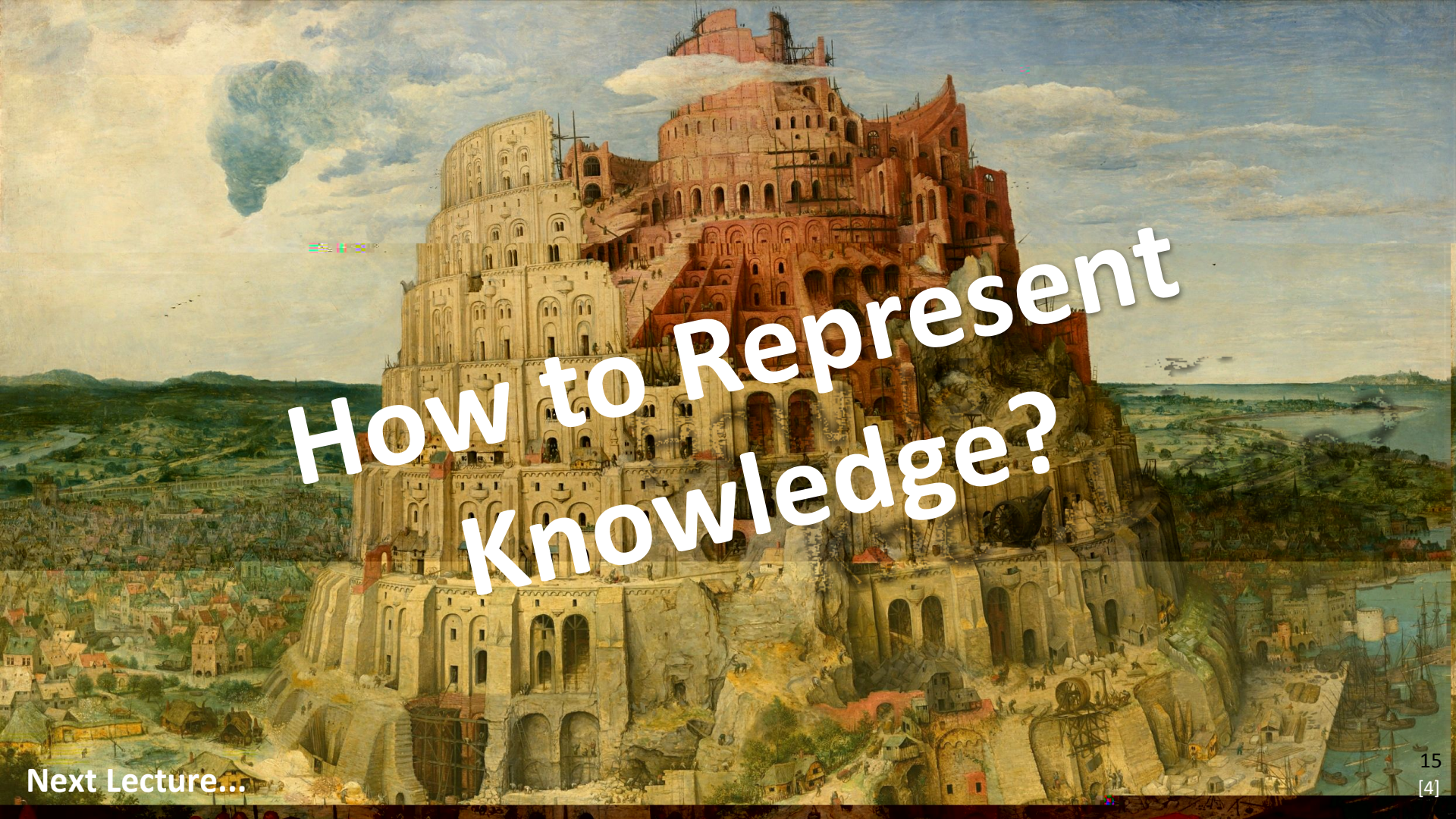
To represent knowledge, we need a **formal knowledge representation = Ontologies**



„People can't share knowledge if
they don't speak a common language“

...to speak a common Language:

- common symbols and concepts (**Syntax**)
- agreement about their meaning (**Semantics**)
- classification of concepts (**Taxonomy**)
- associations and relations of concepts (**Thesauri**)
- rules and knowledge about which relations are allowed and make sense (**Ontologies**)



How to Represent Knowledge?

Next Lecture...

Picture References:

- [1] Matrix Computer Screen, <https://pixabay.com/illustrations/matrix-code-computer-pc-data-356024/>
- [2] UBC Library Card Catalog, Paul Joseph, cc-by-2.0, https://commons.wikimedia.org/wiki/File:2009_3544505541_card_catalog.jpg
- [3] Tree of knowledge based on the French Encyclopedie from 1780, https://commons.wikimedia.org/wiki/File:Essai_d%27une_distribution_g%C3%A9n%C3%A9alogique_des_sciences_et_des_arts_principaux_1780.jpg
- [4] Pieter Bruegel the Elder, The Tower of Babel (1563) [https://commons.wikimedia.org/wiki/File:Pieter_Bruegel_the_Elder_-_The_Tower_of_Babel_\(Vienna\)_-_Google_Art_Project.jpg](https://commons.wikimedia.org/wiki/File:Pieter_Bruegel_the_Elder_-_The_Tower_of_Babel_(Vienna)_-_Google_Art_Project.jpg)